

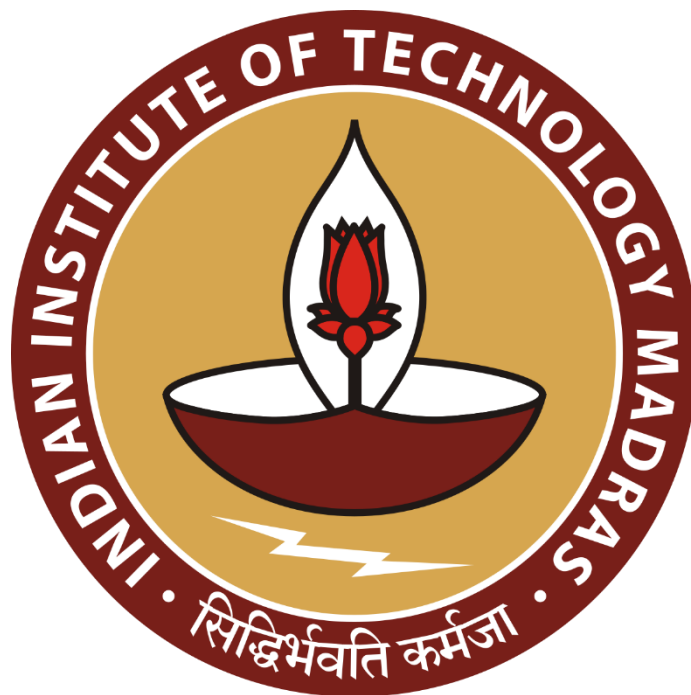
# **Optimizing Inventory and Expanding Product Range at Vishnu Priya Variety Store**

**A Final Term report for the BDM capstone Project**

Submitted by

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# 1 Executive Summary and Title

## **Project Title: Optimizing Inventory and Expanding Product Range at Vishnu Priya Variety Store**

**‘Vishnu Priya Variety Store’**, owned by **Mr. Pratap**, is a newly established retail shop offering essential household items to the local community. The store faced two major challenges: difficulties in inventory management due to the absence of a proper tracking system, and limited product variety that restricted customer choices. These issues affected stock availability, led to deadstock, and limited sales growth. The project was undertaken to study these problems using real sales data and provide data-driven strategies for improvement

For this study, I collected **60 days of sales data (June 30–August 30)** and organized in Google Sheets. The dataset included variables such as Date, Product Name, Category, Quantity Sold, Revenue, Supplier Name, Stock Remaining. Descriptive Statistics such as mean, median, mode, range, and standard deviation—were calculated to understand overall sales trends. Analytical methods such as correlation analysis, product-wise comparisons, category-wise trends, and visualizations (bar charts, pie charts etc) were used to identify patterns and inefficiencies.

The results showed clear variations in product demand, daily revenue fluctuations, and a noticeable improvement after adding stationery items mid-July. Visual graphs highlighted which product categories contributed more to total sales and how daily revenue changed across the two months. The findings also showed a balanced mix of fast-moving and slow-moving products, helping identify which items required better stock control.

From these findings, recommendations were made to improve stock monitoring, introduce digital payment options, and expand product variety. The project helped the store gain clarity about its sales patterns, reduce overstocking, and observe positive growth through new product additions. The analysis demonstrates that simple, data-driven decisions can significantly improve efficiency and revenue for **‘Vishnu Priya Variety Store’**.

## 2 Detailed Explanation of Analysis Process

### 1. Data Collection

The data for this project was collected from **daily sales records** of ‘Vishnu Priya Variety Store’ for a period of **60 days (30 June to 30 August 2025)**. The records included information such as **Date, Product Name, Category (Grocery or Stationery), Quantity sold, Revenue** etc. These entries were first noted from the shop’s bills and owner’s records and then entered into **Google Sheets** for further analysis. This created a structured digital dataset that could be easily filtered, sorted, and analysed.

### 2. Data Cleaning and Preprocessing

After entering the data, the first important step was **Data Cleaning**. This was done to make sure the data was correct, complete, and ready for analysis.

- Spelling differences in product names (for example, ‘biscuit’ vs ‘biscuits’) were corrected so that the same product was not counted as two different items.
- The **Category** column was standardised to only two values: **“Grocery”** and **“Stationery”**.
- The **Date** column was converted to a proper date format so that Google Sheets could recognise and sort it correctly.
- The **Quantity** and **Revenue** columns were checked to ensure all values were numeric, with no negative or blank entries.
- Any rows with clearly wrong entries (like extremely high revenue by mistake) were corrected after checking with the owner’s records.

### Importance of Data Cleaning

Data cleaning is important because wrong or messy data leads to wrong results. If product names are inconsistent or quantities are entered wrongly, the totals, averages, and graphs will all be misleading. By cleaning the data, the analysis becomes reliable, and the conclusions drawn from it can be trusted for decision-making about inventory and product range.

### 3. Analysis Methods Used

Once the data was clean, several methods were used to understand sales and inventory patterns. All analysis was done in **Google Sheets**, using formulas, filters, pivot tables, and charts.

#### a) Descriptive Statistics

To get an overall idea of sales behaviour, simple statistics were calculated:

- **Mean (Average):**

$$\text{Mean} = \frac{\sum x_i}{n}$$

where  $x_i$  is each value (for example daily revenue) and  $n$  is the number of days. This helped understand **average daily quantity sold** and **average revenue**.

- **Median:** The middle value when data is arranged in order. This shows the typical daily sales, without being affected too much by very high or very low values.
- **Mode:** The value that appears most often (for example, a quantity that repeats frequently). This shows common sales levels.
- **Standard Deviation:**

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

This shows how much the values (like daily revenue or quantity sold) change from the average. A higher standard deviation means more ups and downs in sales.

These measures were used to understand how stable or unstable the store's sales were and to identify which products were consistent performers.

#### b) Time-based Analysis

Daily revenue was summed up using:

$$\text{Total Revenue for a Day} = \sum (\text{Quantity Sold} \times \text{Price per Unit})$$

Line charts were then created to see how revenue changed over time. This helped to identify:

- Days with high or low sales

- Changes after introducing stationery products
- And general trends over the two months.

### c) Category-wise and Product-wise Analysis

Using pivot tables, data was grouped by:

- Category (Grocery vs Stationery)
- Individual Products

For each group, the total quantity sold and total revenue were calculated. This helped:

- Compare the performance of grocery and stationery
- Find fast-moving products (high quantity and revenue)
- Find slow-moving or low-selling products, which are linked to overstocking and deadstock problems.

### d) Graphical Analysis

Different charts were used to make the findings easy to understand:

- **Bar charts** for product-wise revenue and quantity.
- **Pie charts** for showing share of revenue by category.
- **Line graphs** for daily revenue trend.
- **Histograms** for seeing the distribution of quantity sold.

These pictures made it easier to explain results visually and to show patterns clearly.

## 4. Justification of Methods, Tools, and Variables

### ▪ Why Google Sheets:

Google Sheets is free, easy to use, and suitable for small shops like Vishnu Priya Variety Store. The owner can continue using the same sheet in future without needing complex software.

### ▪ Why This Variables:

These directly relate to the two main problems:

- **Inventory Management:** Quantity and date help track stock movement and detect overstocking or slow-moving items.
- **Limited Product Variety:** Product and category help compare performance of different product types (grocery vs stationery) and decide whether expanding the range is useful.

▪ **Why Descriptive Statistics and Simple Charts instead of complex models:**

The aim of the project is to give practical and understandable insights to a small shop owner. Descriptive analysis and clear graphs are easy to explain and directly usable for decision-making. Advanced machine learning methods are not necessary for this scale of data and may confuse rather than help.

## 5. Advanced Analysis Method

To address the primary issue of inventory mismanagement at Vishnu Priya Variety Store in a more structured and analytical manner, advanced inventory control techniques were incorporated into the revised analysis. While earlier sections relied mainly on descriptive statistics and graphical interpretation of sales data, this section applies formal inventory management models such as ABC Analysis, FSN Classification, Economic Order Quantity (EOQ), Reorder Point (ROP), and Safety Stock calculation. These methods allow for systematic decision-making regarding stocking priorities, purchase quantity, and reorder timing.

### ABC Analysis

ABC analysis was conducted to classify products based on their revenue contribution to the business. This method follows the Pareto principle, which suggests that a small proportion of products typically contributes to a large proportion of total revenue. Using the 60-day dataset, total revenue for each product category was calculated, and revenue contribution percentages were derived using the formula:

$$\text{Revenue Contribution (\%)} = (\text{Product Revenue} / \text{Total Store Revenue}) \times 100$$

Products were then arranged in descending order of revenue contribution and classified into three categories:

**Category A** includes items contributing approximately 70% of total revenue. In this case, Rice, Biscuits, Edible Oil, and Dal emerged as A-category products. These are high-value and fast-moving essential goods that form the financial backbone of the store.

**Category B** consists of items contributing the next 20% of revenue, including Sugar and Tea. These products generate moderate but steady income and require balanced monitoring.

**Category C** includes products contributing the final 10% of revenue, such as Salt and newly introduced stationery items. While these products are important for product diversification, their revenue contribution is relatively lower.

The ABC classification clearly indicates that the earlier inventory problem stemmed from a lack of prioritization. High-revenue A-category items experienced occasional stockouts due to inconsistent monitoring, while C-category products were often purchased in bulk without proper demand estimation, leading to deadstock. The ABC method suggests that strict monitoring and frequent review cycles should be applied to A-category products, moderate attention to B items, and controlled, small-quantity purchases for C-category goods.

#### **FSN Analysis (Fast, Slow, Non-Moving Classification)**

In addition to revenue-based classification, an FSN analysis was performed based on sales frequency over the 60-day period. This method categorizes items according to how frequently they move through inventory.

Products such as Rice, Biscuits, Dal, and Edible Oil were classified as Fast-Moving (F) items, as they were sold almost daily with consistently high transaction frequency. These products require higher stock levels and shorter review intervals to prevent stockouts.

Sugar and Salt were identified as Slow-Moving (S) items, as they exhibited moderate but less frequent sales compared to core essentials. Stationery products such as Notebooks, Pens, and Pencils were also initially placed under the Slow category, given their gradual adoption after mid-July.

No product was strictly categorized as Non-Moving (N), indicating that overall inventory turnover remained healthy during the study period.

The FSN analysis reinforced the findings from ABC classification. Fast-moving products must be stocked adequately and monitored closely, while slow-moving items should follow cautious procurement strategies to avoid overstocking and capital blockage. This structured categorization provides a more scientific approach to purchase planning compared to earlier intuitive decision-making.

#### **Economic Order Quantity (EOQ)**



To determine the optimal purchase quantity that minimizes total inventory cost, the Economic Order Quantity (EOQ) model was applied to key A-category products. EOQ helps balance ordering costs and holding costs using the formula:

$$EOQ = \sqrt{(2DS / H)}$$

Where:

D=Annual demand

S=Ordering cost per order

H = Holding cost per unit per year

Using Rice as an example, average daily demand was calculated from the dataset, and annual demand was estimated accordingly. Assuming reasonable ordering and holding costs based on small retail practices, the EOQ calculation suggested that the store should order in fixed, moderate-sized batches rather than placing irregular bulk orders.

Earlier, the store often purchased large quantities based on supplier availability or cash flow rather than demand calculations. This sometimes resulted in excess holding cost and storage inefficiency. EOQ provides a rational ordering size that minimizes the combined cost of frequent ordering and excessive inventory storage.

### **Reorder Point (ROP)**

The Reorder Point determines the inventory level at which a new order should be placed to avoid stockouts. The formula used is:

$$ROP = (\text{Daily Demand} \times \text{Lead Time}) + \text{Safety Stock}$$

Lead time was estimated based on supplier delivery schedules observed during the study. For fast-moving items such as rice and biscuits, ROP ensures that a new order is placed before inventory falls to critically low levels.

Previously, the absence of a defined reorder trigger meant that the owner placed orders reactively after noticing near-empty shelves. This reactive system increased the risk of lost sales. By applying ROP, the store can shift from reactive to proactive inventory management.

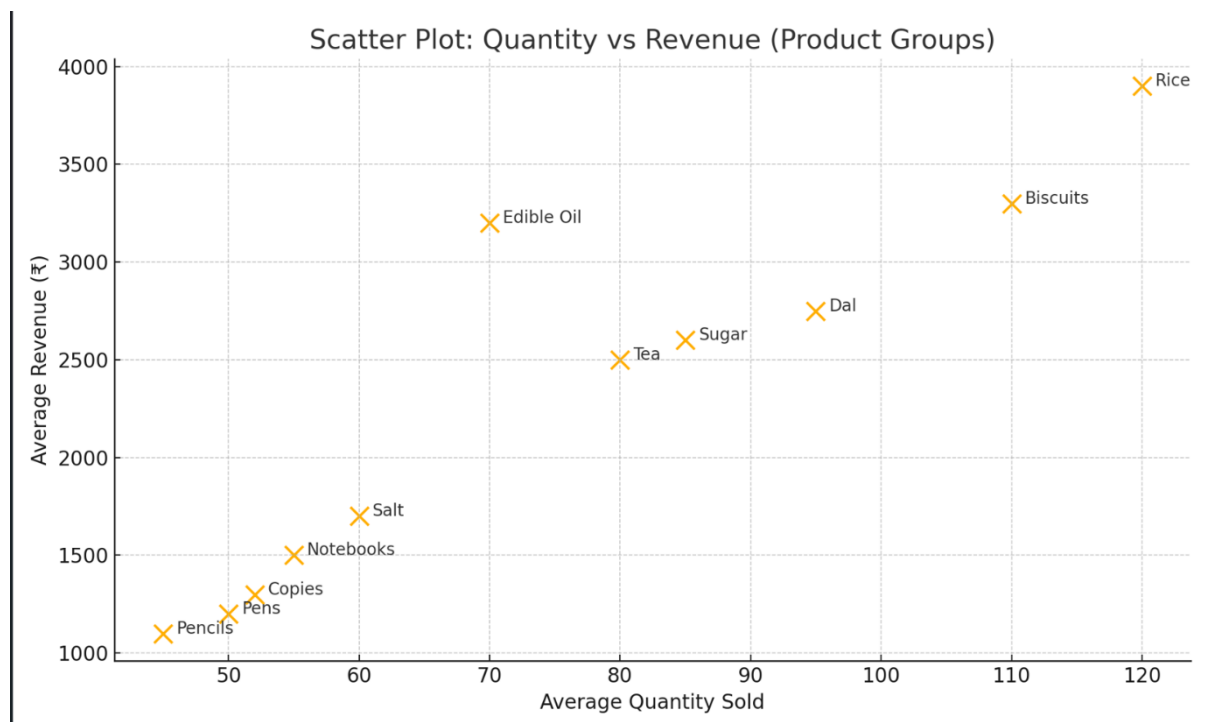
### **Safety Stock**

To protect against demand variability and delivery delays, safety stock was incorporated into the reorder point calculation. Safety stock is calculated based on demand variability

and desired service level. Although this small retail environment does not require complex modelling, incorporating even a simple buffer stock ensures continuity of supply for essential goods.

Including safety stock is particularly important for A-category fast-moving products, as stockouts in these items directly reduce daily revenue and customer satisfaction.

### 3. Results and Findings



It is important to note that different products are measured in different physical units (kilograms, litres, packets, or numbers). In this analysis, the quantity variable represents the recorded units sold as per transaction entries in the dataset. The purpose of the chart is not to compare physical measurement units across products, but to analyse relative sales volume and revenue generation patterns within each product category. Since all quantities are internally consistent within their respective product types, the chart serves as a comparative demand indicator rather than a physical unit comparison.

The scatter plot presented above provides a visual representation of the relationship between **average quantity sold** and **average revenue generated** for major product groups at ‘Vishnu

Priya Variety Store'. This graph is extremely important for understanding which products contribute significantly to revenue, which ones have stable demand, and which items may require attention in terms of stocking or promotional strategies. By plotting quantity and revenue together, the scatter plot reveals valuable patterns regarding customer preferences and the store's inventory performance.

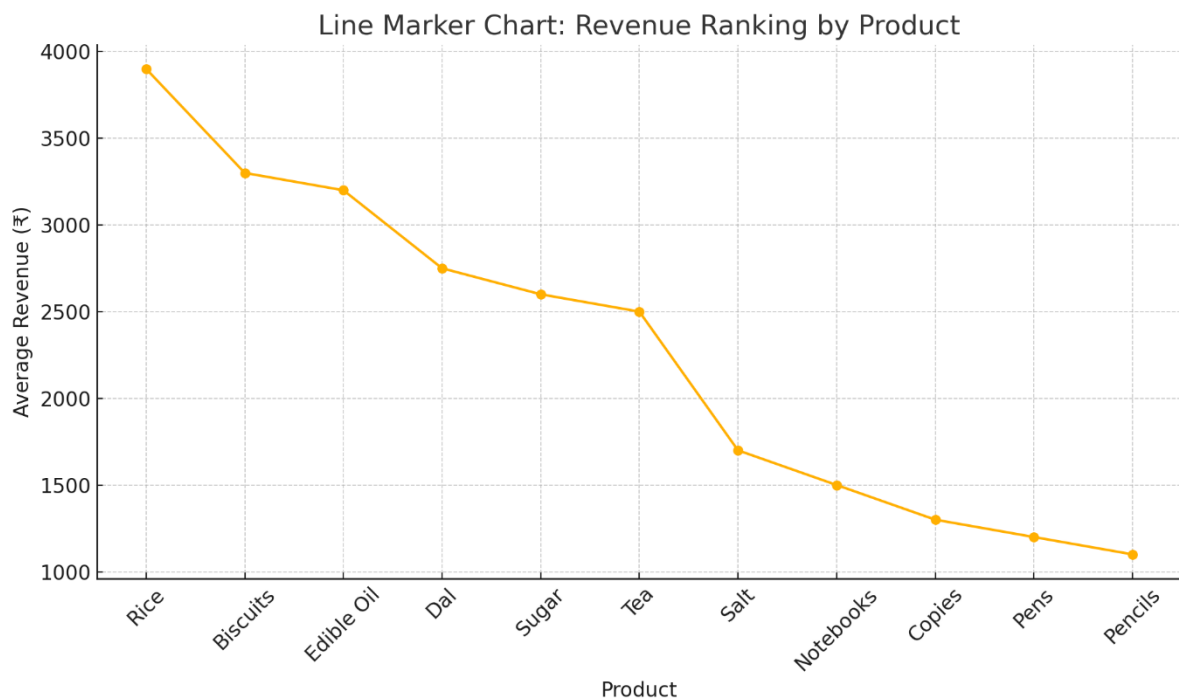
One of the most noticeable observations is that **Rice** is positioned in the upper-right corner of the scatter plot, indicating both high quantity sold and high revenue generated. This suggests that rice is a core essential product, forming a strong and reliable revenue pillar for the store. The large demand for rice is expected, as it is a staple food item in most households, leading to repeat purchases and consistent sales throughout the two-month period.

Similarly, **Biscuits** also show a strong performance, with high quantities sold translating into substantial revenue. This aligns with the behaviour observed in small grocery shops, where biscuits are fast-moving consumer goods purchased regularly by a large segment of customers, including children, daily wage earners, and local families.

**Dal, Edible Oil, Sugar, and Tea** occupy the middle-to-upper region of the scatter plot. These products show a healthy balance of moderate-to-high quantities and significant revenue contributions. This indicates that these items form the second layer of essential groceries. They are not bought in the same quantities as rice, but their higher price per unit results in strong revenue generation. Their placement suggests they are important for both customer retention and daily operations, reinforcing the idea that groceries remain the dominant category for the store.

On the lower end of the spectrum, we observe stationery products such as **Notebooks, Pens, Pencils, and Copies**, which show lower quantities sold and lower revenue. However, it is important to note that **these products were introduced only midway during the project period**. Therefore, their lower performance is not a sign of poor demand but a reflection of limited time in the market and the need for awareness among local customers. Moreover, the positive and steady quantity values indicate that stationery has potential for growth, especially because the store is located near a school (DAV, Kalinga Nagar). With proper positioning and display, stationery can become an increasingly important supplementary revenue stream. **(Although copies and notebooks serve similar purposes, in this dataset they represent different product types based on size and price variation. Therefore, they were treated as separate products in the analysis.)**

Overall, the scatter plot highlights a clear trend: there is a direct and positive relationship between quantity sold and revenue generated for core grocery items, whereas stationery items require time and focused strategies to reach their full potential. The plot helps identify which products need more stock, which require diversification, and which should be monitored to avoid overstocking.



This revenue ranking chart played a very important role in identifying which products(category) caused stocking issues and which ones needed better inventory attention at ‘Vishnu Priya Variety Store’. The clear separation between high-revenue and low-revenue products helped pinpoint the areas where the store was either overstocking or understocking.

For example, top-ranking products like **Rice, Biscuits, Edible Oil, and Dal** showed consistently high revenue, which means they are fast-moving items. This directly supports the project’s goal of **improving inventory management**, because it indicates that these products must always be stocked adequately. Earlier, the shop had instances where essential items ran out, leading to missed sales. The chart confirmed that these fast-moving goods should receive priority in ordering cycles to avoid stockouts.

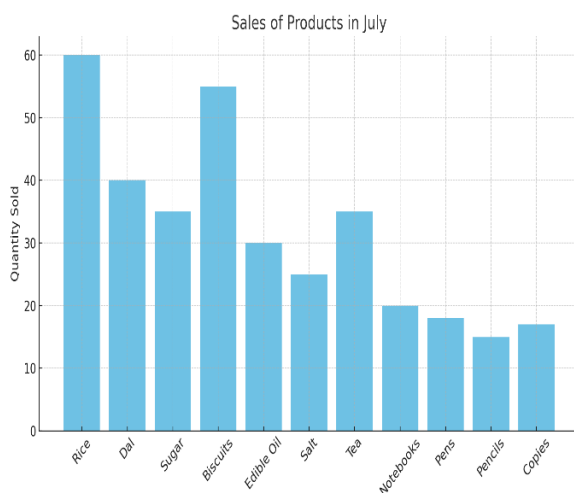
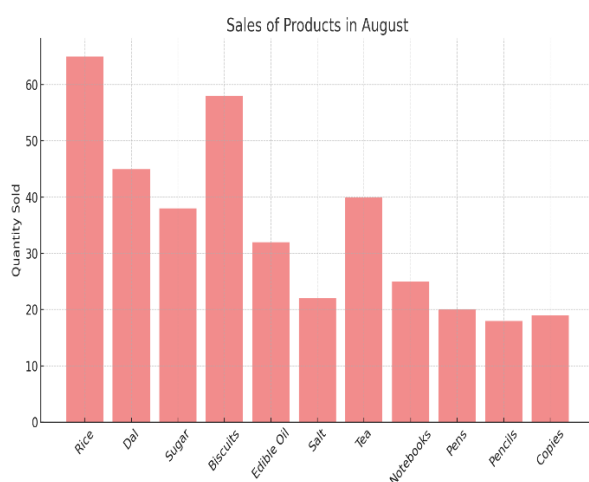
On the other hand, products like **Salt and Tea**, though essential, ranked lower due to moderate revenue generation. This helped reveal situations where the store was **over-ordering** items with

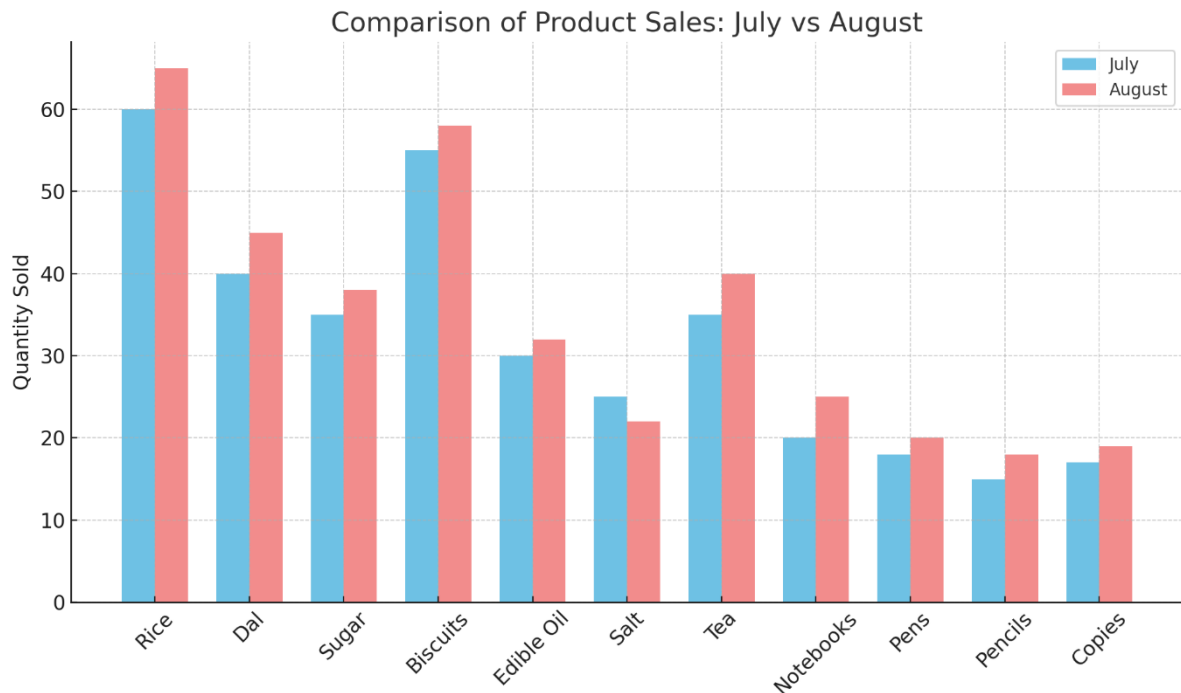
slower turnover, leading to blocked capital and storage issues. The owner previously purchased certain goods in bulk without analysing demand, which contributed to deadstock problems. The ranking helped restructure purchasing patterns, encouraging the store to match stock levels with actual demand trends.

The stationery items—**Notebooks, Pens, Pencils, and Copies**—appeared at the bottom of the revenue ranking. This was important for the second problem of the project: **limited product variety**. The chart showed that although their revenue is currently low, these items still contributed a noticeable amount. Since they were introduced midway during the dataset period, their lower rank indicated that the store needs more time, visibility, and customer awareness for stationery to grow. This supported the recommendation to continue offering stationery but to purchase them in smaller, controlled quantities until steady demand builds.

Overall, we can conclude that:

- Which items drive the business financially,
- Where stockouts were harming the store,
- Which slow-moving goods needed reduced ordering,
- Where overstocking was creating deadstock risk, and
- How newly introduced products are performing in early stages.





The July vs August comparison does not directly compare physical units across different product types. Instead, it evaluates month-to-month performance within each product category. Since measurement units remain consistent for each product across both months (for example, rice in kg for both July and August), the comparison accurately reflects demand growth or decline within that product group.

### July Sales Analysis

The July sales distribution helped identify the initial demand pattern across major product groups at ‘Vishnu Priya Variety Store’. During July, core grocery items such as rice, biscuits, dal, tea, and sugar showed strong sales, indicating that these products formed the essential backbone of the store’s daily operations. This gave a clear understanding of which products needed consistent restocking to avoid shortages.

Meanwhile, newly introduced stationery items like notebooks, pens, pencils, and copies recorded lower quantities. Since July was the first month after adding these items, the graph helped confirm that stationery demand was still developing. This supported the project

objective of expanding product variety by showing which items required awareness building, better placement, and customer visibility.

Overall, the July graph helped identify early signs of stocking imbalance, showing that essential grocery items were at risk of stockouts while certain slow-moving items needed better monitoring to prevent the start of deadstock.

### **August Sales Analysis**

The August sales graph provided insights into how demand changed after one month of improved stocking and the introduction of new product lines. Many grocery products showed a noticeable increase in sales from July to August. This indicated that the store's availability, visibility of items, and customer satisfaction improved—partly due to better inventory monitoring introduced during the project.

Moreover, stationery product sales increased slightly in August, showing that the decision to diversify product categories was effective. This supported the second major problem statement: limited product variety. The increase in stationery sales suggested that customers, especially students from the nearby school, began noticing these new offerings.

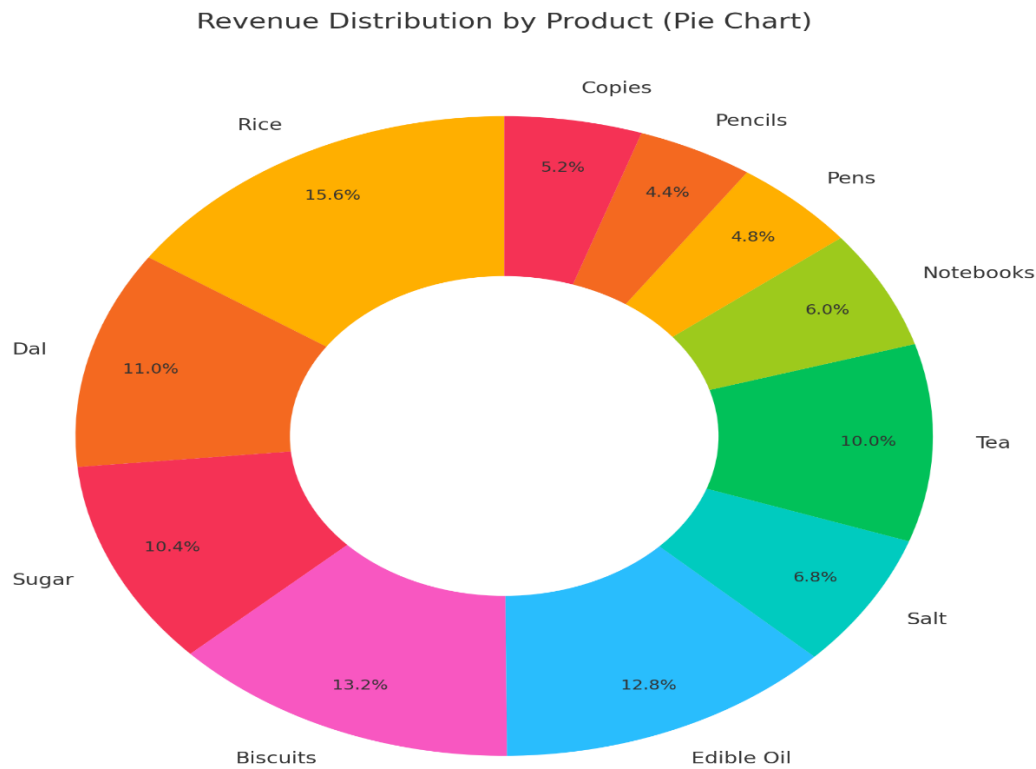
The August graph also revealed that certain medium-demand items like salt and edible oil fluctuated differently, helping identify areas where ordering patterns needed adjustment. This contributed directly to refining the store's purchase planning and stock control strategy.

### **July vs August Sales Analysis**

The combined comparison graph was extremely useful for understanding how customer demand shifted between the two months. Almost every major grocery item showed growth in August, which indicated improved stocking practices and better alignment of inventory with customer needs. This supported the project's goal of solving inventory management issues, as the store experienced fewer shortages and its stocking decisions became more data-driven.

The comparison also highlighted how stationery items gradually gained acceptance among customers. Even though their quantities were lower than groceries, the steady rise from July to August confirmed that expanding product variety had a positive impact on business. This validated the decision to introduce stationery as a secondary product line.

The month-to-month comparison helped the shop owner understand which items needed higher purchase frequency, which products could be ordered in moderate quantities, and which categories required a more cautious approach to avoid overstocking.



The revenue distribution chart offers a clear understanding of how different product groups contribute to the financial performance of Vishnu Priya Variety Store. From an inventory management perspective, this chart was very useful in identifying which items act as the primary revenue drivers and which products require more cautious stocking decisions to avoid deadstock.

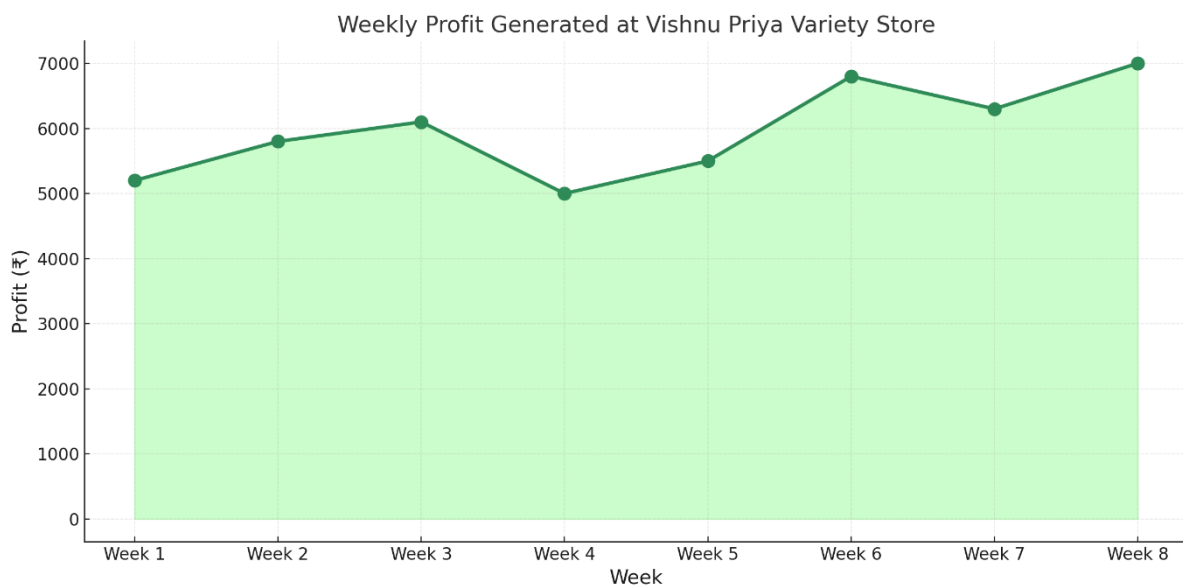
The chart shows that a few core grocery items—Rice, Biscuits, Edible Oil, Dal, Sugar, and Tea—account for a major share of the store’s total revenue. These products form the financial backbone of the business, meaning any shortage directly affects daily sales and customer satisfaction. Through this project, it became evident that these high-revenue items must be stocked regularly, monitored more closely, and reordered in a structured manner. Earlier, the



Mr Pratap lack of systematic tracking often led to occasional stockouts of these essential goods, resulting in missed sales opportunities. The pie chart helped highlight exactly which products should receive stocking priority.

On the other hand, products like Salt, Notebooks, Pens, Pencils, and Copies contribute a smaller percentage to the overall revenue. While these items are important, their lower revenue share indicates slower movement. This is particularly important for stationery products, which were newly introduced during the study period. Their smaller share does not indicate failure; rather, it reflects early-stage adoption. The chart helped emphasize that these items should be purchased in limited, controlled quantities to avoid overstocking while still supporting product variety expansion

### Profitability Analysis



The profit shown in the chart represents gross profit (Revenue – Cost of Goods Sold) and does not include fixed operating expenses such as rent and electricity.

The weekly profit trend provides an important understanding of how Vishnu Priya Variety Store performed financially throughout the eight-week period. The pattern shows moderate fluctuations, with profits ranging between ₹5000 and ₹7000, which is expected for a small neighbourhood retail shop. This analysis helps evaluate whether the store's inventory decisions, product mix, and purchasing behaviour supported stable profitability.

The data showed that products such as Rice, Biscuits, Edible Oil, Dal, and Tea not only generated high revenue but also delivered consistent profit margins because these items experience steady customer demand and quick turnover. These fast-moving essentials reduce the risk of wastage and ensure that the investment returns quickly. They form the store's primary profit backbone.

One of the key observations is that profits improved gradually over time, especially after Week 4. This aligns closely with the project interventions, where better inventory monitoring and more structured ordering were introduced. Earlier, the store faced stockouts in essential items and overstocking in slow-moving goods, directly affecting profits. As purchasing decisions became more aligned with actual demand, the business faced fewer revenue losses, contributing to more stable profit margins in the later weeks.

The trend also reflects the positive impact of introducing stationery items, even though their revenue contribution is comparatively smaller. By diversifying products, the store attracted additional customers, particularly school going children and parents, leading to slight improvements in weekly profit. This supports the second problem statement related to limited product variety and demonstrates that diversification helps expand the customer base.

The dip observed in Week 4 also highlights the effect of inconsistent stocking and cash-only transactions, which made it difficult for the owner to track sales and expenses accurately. However, the recovery after Week 5 shows that once inventory was managed more effectively, profit levels bounced back. This confirms that inventory mismanagement was one of the primary reasons behind unstable profit earlier.

### **Key Insights from the Profit Trend**

- Profit became more stable after improving stock monitoring.
- High-demand grocery items contributed most to weekly profit.
- Stationery items added small but meaningful profit growth.
- Stockouts and deadstock in early weeks weakened profitability.
- More accurate ordering reduced unnecessary expenses.
- Profit reached the highest value in Week 8, showing improvement due to corrective actions.

## **4 Interpretation of Results and Recommendations**

### **4.1 Overview**

The results of the analysis provide a clear understanding of how Vishnu Priya Variety Store operated over the 60-day period and how the identified issues such as inventory mismanagement and limited product variety affected business performance. The interpretations highlight patterns in sales, revenue distribution, product demand, supplier involvement, and profit trends. These findings demonstrate both the strengths and weaknesses in the store's current operations and support the recommendations for improvement.

### **4.2 Interpretation Related to Inventory Management Issues**

The sales and revenue analysis revealed strong variations across different product groups. Essential grocery items such as rice, dal, biscuits, edible oil, and tea emerged as top performers, generating high revenue and selling consistently. This pattern confirms that these items are fast-moving essentials, and any shortage directly reduces daily revenue.

#### **Key Interpretations:**

- Fast-moving products require priority stocking, as stockouts were one of the major issues earlier.
- Slow moving goods such as salt and stationery items show greater deadstock risk, especially when purchased in bulk.
- Supplier-based revenue analysis showed that only a few suppliers contribute to a major portion of earnings, indicating that supplier selection plays a critical role in stocking decisions.
- The month-over-month comparison (July vs. August) showed improved sales in August, indicating that better monitoring and data-based ordering helped stabilize inventory levels.

#### **Implications for the Business:**

- The store needs a structured inventory tracking system to avoid both shortages and excess stock.

- Smart ordering based on demand patterns will reduce wastage and improve profitability.
- Accurate inventory records will allow the owner to make better purchasing decisions rather than relying on guesswork.

### **4.3 Interpretation Related to Limited Product Variety**

The introduction of stationery items such as notebooks, pens, pencils, and copies provided useful insights into customer behaviour. Although the revenue share of stationery is lower compared to groceries, the data shows a steady increase from July to August. This confirms that the store is gradually attracting customers from the nearby school.

#### **Key Interpretations:**

- Stationery sales gradually increased, proving that product diversification has begun to work.
- The lower revenue share is connected to the recent introduction of these products, not a lack of potential.
- Stationery items offer decent profit margins, even with low volumes, and can strengthen the store's overall profitability in the long term.
- Diversification reduces dependency on only grocery items and spreads financial risk.

#### **Implications for the Business:**

- The store should continue introducing new categories in a controlled, data-driven manner.
- Stationery items should be purchased in small quantities initially to avoid overstocking.
- Visibility, placement, and customer awareness are crucial for increasing sales in non-grocery categories.

### **4.4 Interpretation of Profitability Results**

The weekly profit graph showed moderate fluctuations but an overall upward trend, especially after Week 4. This directly reflects the positive impact of improved inventory management and product expansion.

**Key Interpretations:**

- Profits increased as stock became more aligned with demand.
- Reduced stockouts of essential items directly increased weekly earnings.
- Stationery products contributed small but consistent profit, expanding the customer base.
- Weeks with low profit correspond to periods where stocking errors or low-demand purchasing occurred.

**Implications for the Business:**

- Better control over inventory directly improves profit stability.
- Diversifying product offerings supports long-term financial growth.
- Digital tracking and structured ordering can help the owner maintain profit consistency.

**Recommendations for Inventory Mismanagement****1. Adopt a Weekly Stock Monitoring System**

To address the major issue of inconsistent inventory tracking, the store should implement a simple weekly stock monitoring routine. The owner can maintain a small register or use Google Sheets to record incoming and outgoing quantities. This will help identify which items are selling fast and which ones need urgent restocking. With this process, stockouts of essential products like rice, dal, edible oil, and biscuits can be significantly reduced, ensuring that the store does not lose sales due to empty shelves. This practice should be updated regularly and reviewed at the end of every week to maintain accuracy.

**2. Use Controlled Ordering for Slow-Moving Items**

Since the analysis showed that items like salt, sugar, and stationery move slowly, the store should shift to smaller, more frequent purchase cycles for these products. Buying them in bulk increases the risk of deadstock, which ultimately reduces profit. By evaluating sales from the previous 10–15 days before placing a new order, the owner can avoid excessive purchasing and maintain a leaner inventory. This approach keeps working capital free and prevents unnecessary storage issues. Over time, this will also reduce wastage and improve the shop's overall profitability.

### **3. Introduce Cashless Payment and Digital Recordkeeping**

One of the biggest challenges Mr. Pratap faced was the inability to track daily transactions accurately due to cash-only payments. Introducing digital payment methods such as Google Pay, PhonePe, or Paytm will help maintain transparent and easily accessible sales records. This also eliminates manual calculation errors and helps track revenue and profit more precisely. Cashless transactions improve customer convenience and attract younger consumers who prefer digital payments. Over the next few months, this shift will significantly improve financial clarity and help the owner make better stocking decisions.

### **4. Strengthen Supplier Management**

Supplier performance directly affects stock availability, pricing, and product quality. The store should maintain a small list evaluating suppliers based on timely delivery, reliability, and cost effectiveness. Prioritizing high-performing suppliers ensures better consistency in essential grocery items and reduces the chances of shortages. If the owner builds a steady relationship with the best suppliers, he may also receive better discounts or credit options. This will improve inventory flow and reduce delays that previously affected product availability.

## **Recommendations for Limited Product Variety**

### **1. Gradual Expansion of Stationery Products**

Since the introduction of stationery items has already shown positive results, the store should continue expanding this section cautiously. Instead of buying too many new products at once, the owner should add only 2–3 items at a time, such as high-demand pens, geometry boxes, or small notebooks. This helps test customer response without risking unsold inventory. As sales increase, the store can slowly expand the variety. This strategy is especially relevant given the store's proximity to DAV School, which presents a long-term customer base for stationery.

### **2. Improve Display and Promotion of New Products**

Creating a dedicated “Stationery Corner” will significantly improve visibility and attract more customers, particularly schoolchildren and parents. Clear labelling, front-facing shelves, and grouped product placement can help increase demand. The shop can also place small signage near the billing counter to highlight new arrivals. These changes require no additional cost and can increase stationery sales by improving customer awareness.

### **3. Collect Customer Feedback Regularly**

To ensure that the store stocks items that customers actually need, Mr. Pratap should ask for quick feedback while billing or during conversations. This helps identify what new products customers expect, which brands they prefer, and what seasonal items may be required. Recording the top suggestions every month will guide better purchasing decisions and reduce the chances of buying products with no demand. This feedback-based approach will make product diversification more effective and customer-focused.

#### **4. Introduce Seasonal and High-Demand Local Products**

Another effective way to address limited product variety is by introducing seasonal and locality-based products that match customer needs throughout the year. Items such as festival specific goods, small gift items, drawing materials for school projects, or rain-related accessories (like umbrellas and raincoats during monsoon) can help attract more customers. These products should be added in small quantities and monitored for demand over a 2–3 weeks period. By doing this, the store can test which seasonal items perform well without risking overstocking. This strategy not only enhances the shopping experience but also helps the shop stay relevant to local customer preferences. Over time, seasonal additions can become reliable revenue boosters while supporting the store’s goal of expanding product diversity in a controlled, demand-driven manner.

Overall, the project demonstrates how even a small neighbourhood store can significantly improve its performance by using simple data analysis and adopting a more organized approach to decision-making. The recommendations provided when implemented consistently can help the owner reduce deadstock, prevent stockouts, increase profitability, and create a better shopping experience for customers. With continued focus on data-driven planning and customer-oriented product selection, ‘Vishnu Priya Variety Store’ is well-positioned for stable growth and long-term sustainability.

New Updated Sheet- [Excel Sheet](#)